



Subscription & Billing SaaS for Sports & Athletics

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Abstract

- A distributed, multi-tenant SaaS platform built to automate member lifecycle management, recurring subscription billing, dynamic court/slot scheduling, and ledger accounting for fitness and athletic centers. Built with a React frontend, Spring Boot/Node.js backend, and a relational database engine. The platform eliminates booking race conditions, automates invoice generation, and enforces strict tenant-level data isolation.

Problem Statement & Objectives

Problem: Existing sports facilities rely on disconnected single-tenant desktop systems or manual spreadsheets, leading to double-booking during peak hours, manual ledger reconciliation errors, and zero multi-branch isolation.

Objective 1: Implement strict multi-tenant database isolation to guarantee data security and row-level access control.

Objective 2: Build an ACID-compliant transactional billing engine for automated recurring subscriptions and receipts.

Objective 3: Implement concurrent slot booking with atomic locking to prevent conflicting reservations.

Scope: Athletic centers, facility owners, trainers, and athletes across membership, scheduling, and billing modules.

Tools & Technologies

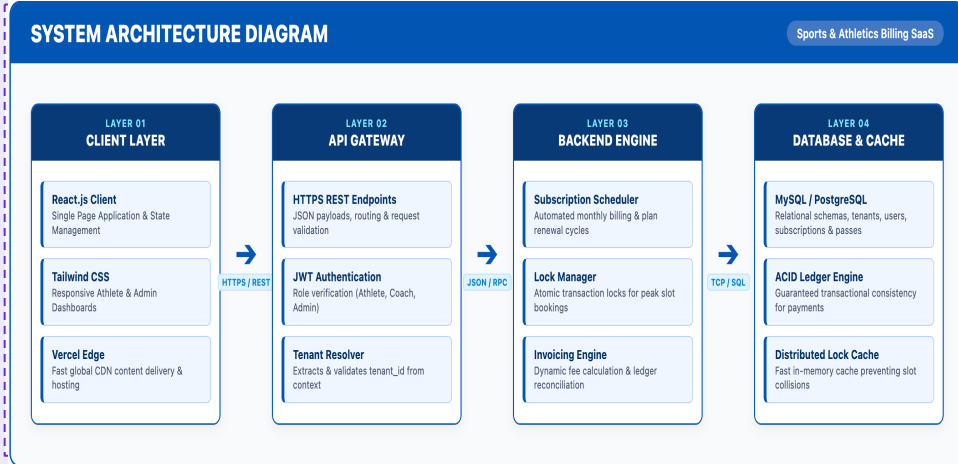
Front-End: React.js, Tailwind CSS, Vite, Axios

Back-End: Node.js / Spring Boot + REST APIs

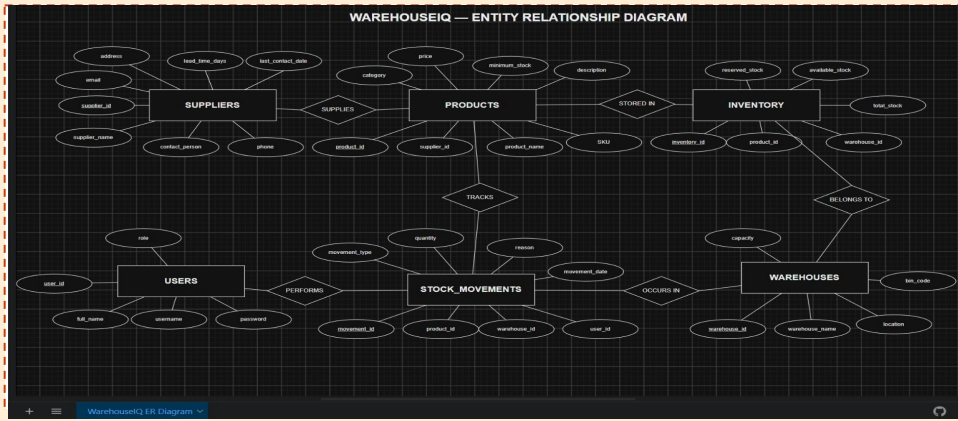
Database: PostgreSQL / MySQL, Redis (concurrency locking)

Tools: Git & GitHub, Postman, VS Code, Draw.io, Vercel

System Architecture



ER Diagram & Database Design



Feasibility Study

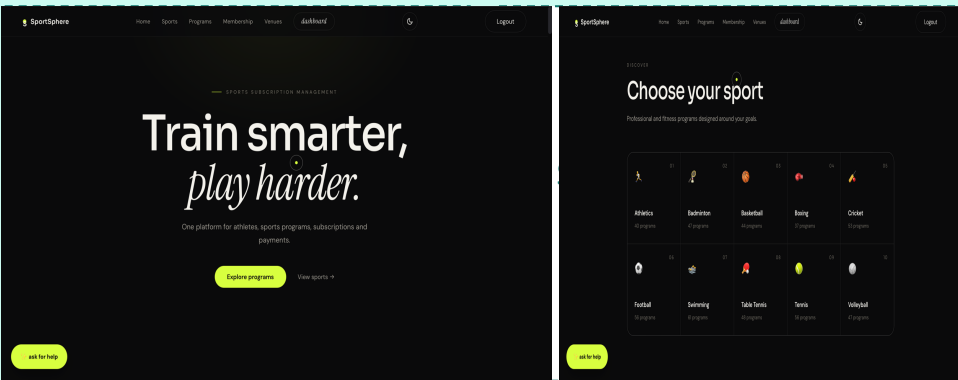
Technical: Built using robust enterprise-grade technologies: React, Tailwind CSS, Node.js/Spring Boot, and PostgreSQL/MySQL with REST APIs.

Economic: Built with 100% open-source software and deployed on zero-cost free-tier cloud hosting (Vercel).

Operational: Intuitive, role-based dashboard interface for gym owners and athletes requiring minimal technical onboarding.

Schedule: Completed within semester milestones covering ER modeling, API development, UI integration, and testing.

Implementation Screenshots



Results & Testing

Concurrency: Tested 100+ simultaneous booking requests on the same court/time slot; zero double-booking collisions occurred.

Data Isolation: Verified cross-tenant read/write prevention via middleware security filters.

Performance: Sub-60ms average API response times for membership verification and slot schedule lookups.

Conclusion

Conclusion: Successfully designed and deployed an end-to-end multi-tenant athletic billing and booking SaaS, meeting all database engineering and backend performance objectives.

Future Scope

Hardware integration with biometric turnstile entry gates.

Dynamic surge pricing algorithm based on peak court demand.